Regularisation in ML

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Introduction

Regularisation adds a complexity penalty to the training loss, preventing overfitting by shrinking coefficients. Ridge (L2) shrinks all coefficients toward zero; lasso (L1) also performs variable selection; elastic net combines both. All three are standard tools across regression and classification.

Prerequisites

Linear / logistic regression; convex optimisation.

Theory

Objective: $\min_\beta \sum_i L(y_i, x_i^\top \beta) + \lambda \cdot P(\beta)$, with - $P(\beta) = \|\beta\|^2$ (ridge, L2) - $P(\beta) = \|\beta\|_1$ (lasso, L1) - $P(\beta) = \alpha\|\beta\|_1 + (1-\alpha)\|\beta\|^2/2$ (elastic net)

Lasso produces sparse solutions (many $\hat{\beta}_j = 0$); ridge shrinks but rarely zeros; elastic net balances sparsity with the grouping effect for correlated features.

Assumptions

Features are standardised so the penalty is comparable across predictors; λ is tuned by CV.

R Implementation

```r
library(glmnet)

set.seed(2026)
n <- 300; p <- 50
X <- matrix(rnorm(n * p), n, p)
beta <- c(rep(2, 5), rep(0, 45))
y <- X %*% beta + rnorm(n)

cv_ridge <- cv.glmnet(X, y, alpha = 0,   nfolds = 10)   # L2
cv_lasso <- cv.glmnet(X, y, alpha = 1,   nfolds = 10)   # L1
cv_enet  <- cv.glmnet(X, y, alpha = 0.5, nfolds = 10)

nz <- function(fit) sum(as.vector(coef(fit, s = "lambda.min")) != 0) - 1
cat("nonzero: ridge=", nz(cv_ridge),
    "  lasso=", nz(cv_lasso),
    "  elastic=", nz(cv_enet), "\n")
c(ridge = cv_ridge$cvm[cv_ridge$lambda == cv_ridge$lambda.min],
  lasso = cv_lasso$cvm[cv_lasso$lambda == cv_lasso$lambda.min],
  enet  = cv_enet $cvm[cv_enet $lambda == cv_enet $lambda.min])
```

Output & Results

Non-zero counts: ridge retains all 50, lasso selects ~5, elastic net selects ~7. Elastic net combines sparsity (lasso-like) with stability (ridge-like) for correlated features.

Interpretation

"Lasso with CV-selected λ retained the 5 true non-zero features and zero-ed out 45 irrelevant ones; elastic net gave similar sparsity but handled correlated predictors more gracefully."

Practical Tips

- Always standardise features (`glmnet` does this internally); otherwise the penalty is scale-dependent.
- Use `lambda.1se` for a sparser, more parsimonious model within one SE of the CV minimum.
- Lasso picks one from correlated-feature groups arbitrarily; elastic net picks all – often more stable.
- For classification, extend via `family = "binomial"`; same rules apply.
- Gradient-boosting frameworks expose L1/L2 regularisation on leaf weights; not identical to glmnet's lasso/ridge but same motivation.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis